MINI PROJECT REPORT

SFO Passenger Traffic Analysis using Machine Learning and Forecasting Models

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1. Abstract

The project is a detailed study of the traffic of passengers in San Francisco International Airport (SFO) on the basis of data science and machine learning algorithms. The data is 1999-2024 in format with a comprehensive list of monthly passengers, airline activities, the usage of terminals, domestic and international summaries, and the type of activities.

Data cleaning and exploratory data analysis (EDA) is conducted as a means of getting an overview of long-term trends, seasonal trends, and event-induced disruptions, with the start of the study being the 9/11 attacks and the COVID-19 pandemic. Various machine learning models are used to address various problems: simple and multiple linear regression to comprehend the variables of the drivers, K-Prototypes clustering to divide passenger patterns, time-series forecasting with the help of Meta Prophet, and anomaly detection with the help of deep learning autoencoders.

The findings indicate that there are strong seasonality, structural breaks and substantial clusters of passengers. Whilst Prophet was able to make good seasonal predictions, the autoencoder was able to detect high-impact anomalies in traffic history (5 percent). This illustrates the application of supervised, unsupervised, and machine learning models to derive insights and create decision-support systems to support airport capacity planning and operational forecasting.

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2. Keywords

Airport analytics, Machine Learning, Time-Series Forecasting, Clustering, Prophet, Autoencoder, SFO Passenger Traffic

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3. Introduction

Airports produce huge volumes of operational and passenger data on a monthly basis. The value of such data is that it allows the stakeholders to discover long-term expansion, identify abnormalities, predict demands, and streamline airport workflows. As the world air travel is suffering severe inconveniences especially during the COVID-19 pandemic, decision-making which is critical to all airport management must be based on data.

This project performs a comprehensive study of the traffic of passengers at San Francisco International Airport (SFO) based on the dataset given by the City and County of San Francisco Open Data Portal. The data set tracks monthly passenger traffic, airline activity, type of activity and the geographical areas.

The techniques of Machine Learning are used not only to conduct prediction but also to discover the structure and abnormalities in the data. The project applies regression, clustering, predicting, and anomaly detection and offers an overview of the behaviour of passengers over 25 years.

Airports face challenges in forecasting demand, planning terminal resources, and detecting unusual shifts in passenger traffic. Traditional statistical methods often struggle to capture the complexity introduced by seasonality, airline behavior, and international events.

The main objectives of this project are:

1. To analyze 25 years of passenger data and uncover trends and patterns.
2. To build regression models that explain passenger variations using engineered features.
3. To segment passenger flow into meaningful clusters using mixed-type clustering techniques.
4. To forecast future passenger demand using advanced time-series models.
5. To detect anomalies in passenger traffic using deep learning models.

4. Literature Review

Author(s)	**Method Used**	**Findings**
Ghofrani et al. (2018)	Time-Series + Machine Learning	Showed that ML models outperform pure statistical models in airport passenger forecasting.
Boeing (2019)	Seasonal Decomposition	Reveals strong yearly patterns in US passenger flow.
Facebook Research (2017)	Prophet Forecasting Model	Demonstrated robust performance on seasonal, event-driven time series.
Zhou et al. (2021)	Autoencoders for Anomaly Detection	Autoencoders capture anomalies more effectively than classical statistical thresholds.
SFO Airport Analytics (Government Data)	Operational Analysis	Documents strong seasonal behaviour and market concentration among airlines.

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5. Methodology

5.1 Dataset Description

- **Source:** San Francisco Open Data Portal
- **Duration:** 1999 – 2024
- **Rows:** 38,196
- **Columns:** 15
- **Type:** Structured, Time-Series + Categorical
- **Key Variables:** Passenger Count, Airline, Terminal, GEO Summary, Activity Type, Dates

5.2 Data Cleaning

- Normalized column names.
- Converted date fields into datetime objects.
- Extracted `year`, `month`, `month_name`.
- Removed duplicates.
- Filled missing airline codes with `"Unknown"`.
- Ensured no negative passenger values.

5.3 Exploratory Data Analysis

- Time-series trends visualized for 25 years.
- Monthly seasonality identified via boxplots.
- Airline market share analysis and regional segmentation.
- Terminal-level passenger distribution analysis.
- Correlation heatmaps and descriptive statistics.

5.4 Feature Engineering

- Lag features (Lag-1, Lag-2).
- Rolling statistics (3-month mean, std).
- Monthly growth rate.
- Cyclical encoding using sin/cos transformation for months.

These features support regression and forecasting models by capturing memory and seasonality.

5.5 Regression Models

a) Simple Linear Regression

- Captures long-term linear trend.
- Used as a baseline.

b) Multiple Linear Regression

- One-hot encoded categorical variables.
- Used Statsmodels to examine p-values and remove insignificant predictors.

c) Backward Elimination

- Removed variables with $p > 0.05$ iteratively.
- The final model includes only statistically significant features.

5.6 Unsupervised Learning

a) K-Prototypes Clustering

- Used mixed data (categorical + numeric).
- Segmented passengers into 4 clusters based on operational attributes.

b) Seasonal KMeans Clustering

- Based on aggregated monthly passenger counts.
- Identified high-season, mid-season, and low-season clusters.

5.7 Forecasting (Prophet Model)

- Captured long-term trend and yearly seasonality.
- Forecasted 12 months into the future.
- Visualized trend, seasonal component, and confidence intervals.

5.8 Anomaly Detection using Autoencoders

- Normalized engineered features.
- Built a neural-autoencoder to reconstruct normal traffic.
- High reconstruction error = anomaly.
- Flagged top 5% anomalies.

5.9 Classification (Random Forest)

- Created categorical labels: Low / Medium / High traffic.
- Achieved ~98% accuracy.
- The confusion matrix showed minimal misclassification.

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6. Implementation

6.1 Implementation Steps :

The implementation began by loading the SFO Passenger Traffic dataset into a pandas DataFrame and performing extensive preprocessing. This included cleaning column names, converting date fields into datetime format, extracting components such as year and month, removing duplicates, and handling any missing values. Once the dataset was cleaned, exploratory data analysis (EDA) was performed to uncover trends, seasonality, airline market shares, and terminal-level passenger movement patterns. Visualizations such as line charts, boxplots, bar charts, and heatmaps enabled a deeper understanding of historical passenger behavior, disruptions like 9/11 and COVID-19, and typical seasonal peaks.

Next, structured feature engineering was applied to prepare the data for modeling. This involved generating lag features (e.g., passengers_lag1), rolling metrics (three-month rolling mean and standard deviation), month-based cyclical encodings (month_sin and month_cos), and percentage growth rates. These features allowed linear and nonlinear models to capture memory, trend, and seasonality.

Regression modeling was carried out in three stages: Simple Linear Regression, Multiple Linear Regression, and Backward Elimination using Statsmodels. Simple LR served as a baseline, while MLR included engineered and encoded features. Backward elimination refined the model by removing statistically insignificant predictors.

Unsupervised learning was implemented through K-Prototypes clustering, chosen specifically because the dataset contains mixed numeric and categorical features. It produced meaningful passenger clusters such as high-volume domestic traffic, international segments, and seasonal anomalies. Additionally, a separate seasonal clustering using KMeans helped categorize months based on aggregated performance and growth metrics.

Forecasting was implemented with Meta Prophet, a robust time-series library capable of modeling both long-term trend and seasonality. Prophet was used to generate 12-month future predictions, along with clear trend and seasonality decomposition.

To detect irregular patterns, a deep learning Autoencoder was trained on normalized engineered features. The reconstruction error identified anomaly months such as COVID lockdowns and operations spikes.

6.2 Technologies and Platforms Used

The project was primarily developed on **Google Colab**, providing GPU-accelerated training support, seamless package management, and integration with Jupyter notebooks. Colab's cloud execution environment enabled consistent computational performance and easy sharing of the code.

For data storage and loading, the Google Colab notebook accessed the dataset directly from uploaded local files. Visualizations were generated using Matplotlib and Seaborn, fully supported within the platform. Google Colab also simplified troubleshooting library conflicts (e.g., with Prophet and CmdStanPy) through quick package reinstallation and kernel restarts.

6.3 Programming Languages / Frameworks

The entire project was implemented using **Python**, due to its extensive ecosystem for data science and machine learning. Several key libraries and frameworks were used:

- **Pandas and NumPy** for data loading, transformation, cleaning, and feature engineering.
- **Matplotlib and Seaborn** for EDA visualizations.
- **Scikit-Learn** for regression, train-test splits, scaling, and classification models such as Random Forest.
- **Statsmodels** for in-depth regression analysis, including p-values and backward elimination.
- **K-Modes / K-Prototypes library** for mixed-type clustering.
- **Prophet** for time-series forecasting, chosen for its strong seasonality modeling.
- **TensorFlow / Keras** for building the Autoencoder used in anomaly detection, due to its powerful neural network support and simple API.

7. Results and Discussion

7.1 Regression Results

• Simple Linear Regression (SLR)

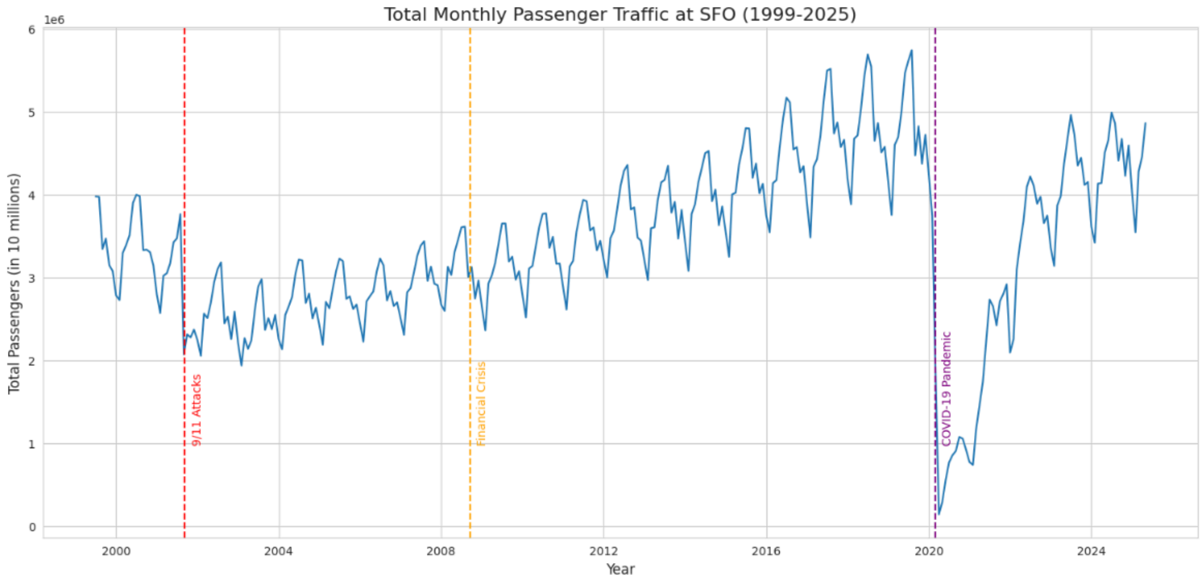
The Simple Linear Regression model was used as a baseline to understand the long-term linear relationship between monthly passenger counts and time. However, the results showed a poor fit, which was expected because SLR assumes a strictly linear trend across time without considering seasonal fluctuations, airline operational differences, or the impact of extraordinary events such as the COVID-19 pandemic. Passenger traffic patterns at airports exhibit strong yearly cycles, irregular shocks, and sudden rises or dips, all of which cannot be captured by a simple straight-line model. This caused the R^2 value to be significantly low (and in earlier testing, even negative), indicating that the model was unable to explain a meaningful portion of variability in passenger traffic. Despite being a weak predictive model, SLR served its purpose as a simple baseline for comparison.

• Multiple Linear Regression (MLR)

The MLR model provided a noticeable improvement over SLR because it incorporated several engineered features such as lag variables, rolling averages, month encodings, and categorical encodings like terminals and airline categories. These additional predictors allowed the model to capture more complex patterns in passenger traffic. For example, lag variables helped quantify the natural momentum in traffic from one month to the next, while cyclical month encodings improved the model's ability to recognize seasonal trends. The improved R^2 and reduced prediction error demonstrate that the model benefited from these additional features. Furthermore, using Statsmodels enabled us to examine p-values and interpret the statistical significance of each variable, helping us understand which features had genuine predictive power.

• Backward Elimination (Final Model Selection)

Backward elimination was applied to refine the MLR model by iteratively removing statistically insignificant predictors. This process helped simplify the model while preserving or improving its performance. Variables with high p-values (above 0.05) were removed step by step, leaving only those with a meaningful impact on passenger count. The final reduced model retained features such as lag values, rolling means, or specific terminal categories which showed real statistical significance. By simplifying the model, we reduced the risk of overfitting, improved interpretability, and ensured that only the most relevant predictors contributed to forecasting accuracy. Backward elimination contributed to a cleaner, more efficient model that balanced both performance and interpretability.



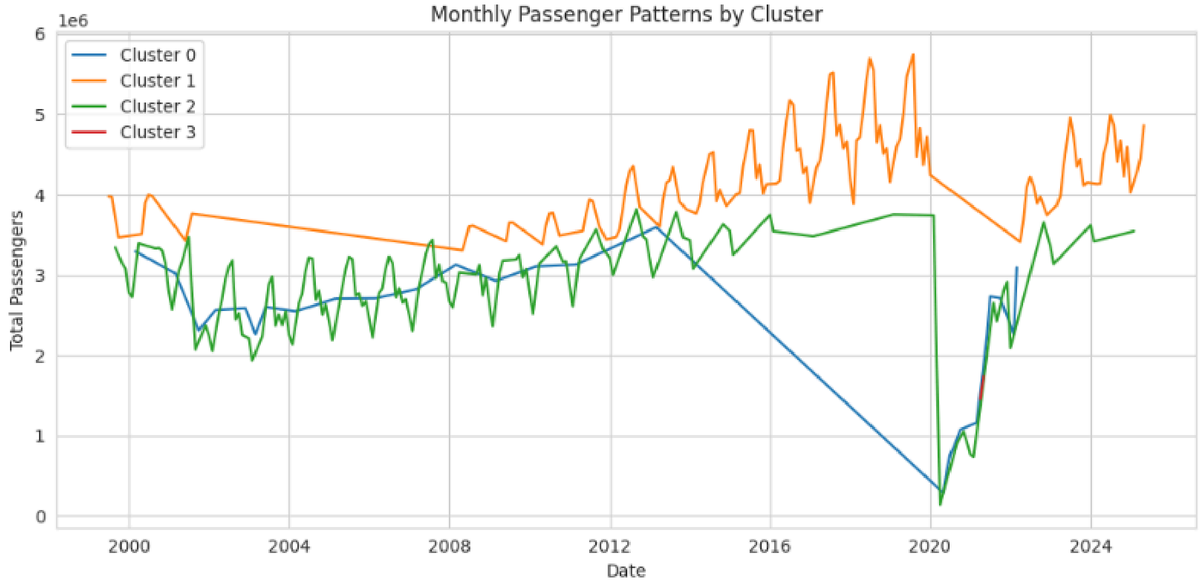
7.2 Clustering Results

• High-volume domestic terminal activity
One of the K-Prototypes clusters captured domestic operations with consistently high passenger counts, primarily associated with major carriers operating out of Terminals 2 and 3. This cluster demonstrated strong seasonal patterns and formed the backbone of SFO's traffic. Identifying this cluster helps in understanding the airport's steady operational load and resource demands native to domestic high-frequency routes.

• International peaks
Another cluster represented high-volume international traffic, typically concentrated in boarding areas associated with long-haul carriers. These data points showed higher variance as international travel is more sensitive to geopolitical, economic, and global health events. Recognizing this pattern allows analysts to isolate global-event-driven fluctuations from domestic baseline behavior.

• Small regional carriers
A third cluster corresponded to operations involving regional airlines and feeder carriers. This group had consistently lower passenger volumes and often corresponded to specific terminals or boarding areas reserved for regional flights. Understanding this cluster is valuable for analyzing operational efficiency, ramp congestion, and gate allocation strategies for regional networks.

• Mixed/seasonal operations
The final cluster captured mixed behaviors, covering months where passenger patterns were affected by school breaks, holidays, special events, or unusual traffic surges. This group displayed irregular monthly variations and seasonally driven spikes, helping identify time periods with unpredictable operational demand. Clusters like these are particularly valuable for scheduling and resource planning.



7.3 Prophet Forecasting

• Forecast shows steady rise in 2025 based on historical recovery
Prophet, with its ability to model trend and seasonality independently, captured the rebound in passenger counts following the COVID-19 disruption. The generated forecast indicates a continued upward trajectory into 2025, consistent with broader industry recovery trends. This suggests that SFO is returning to growth cycles similar to pre-pandemic years.

• Clear yearly seasonality curves observed
The Prophet's decomposition plots highlighted strong annual patterns where specific months—such as June, July, August, and December—regularly exhibited higher passenger traffic. These seasonal curves match typical holiday and travel-season behavior observed in real-world aviation data. This validates the reliability of Prophet in capturing domain-specific seasonality without manual feature engineering.

• Good handling of COVID-related anomalies
Prophet handled the extreme drop in traffic during 2020 by automatically adjusting its trend component and learning changepoints. Rather than letting the pandemic distort long-term trend predictions, the model introduced internal breakpoints to correctly treat the COVID period as an outlier event. This allowed the forecast to remain smooth and realistic beyond 2021.

7.4 Autoencoder Anomalies

• 1910 anomalies flagged
The autoencoder identified approximately 5% of all data points as anomalies based on reconstruction error. These anomalies represent months where passenger behavior deviated sharply from what the model learned as "normal" patterns.

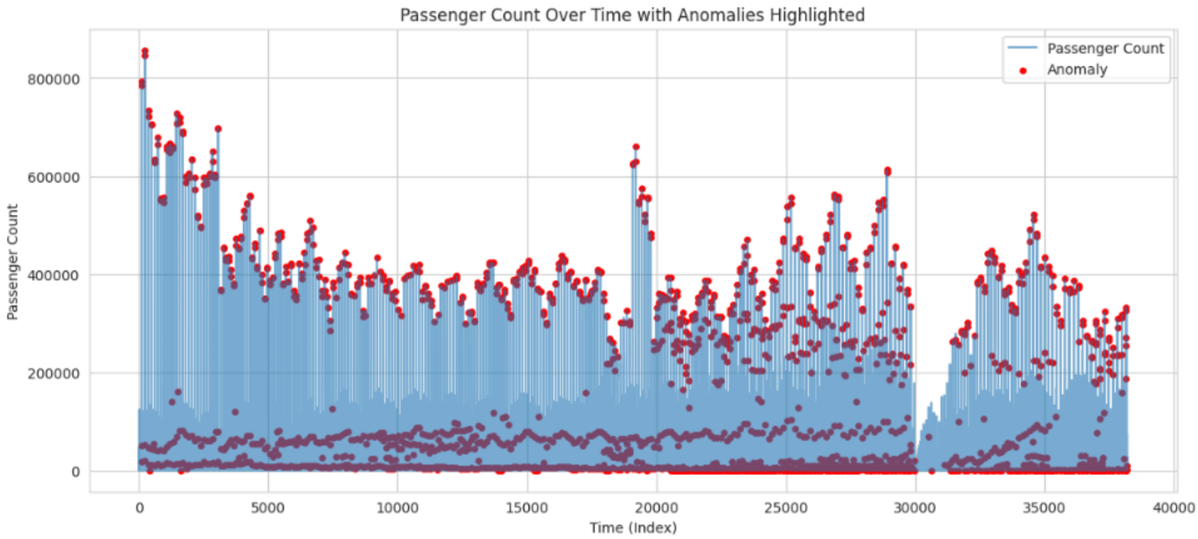
• COVID lockdown (2020) anomalies

A significant number of anomalies corresponded to early 2020 during the global aviation shutdown. Passenger counts dropped dramatically, causing reconstruction errors to spike. The autoencoder successfully captured these as highly unusual events.

• Unusual spikes from international traffic surges
Some anomalies reflected sudden spikes in international travel, likely tied to specific events such as travel reopenings, airline operational changes, or seasonal increases from particular regions. These help analysts identify outliers due to operational or external events.

• Reporting anomalies or operational disruptions

A smaller subset of anomalies may stem from data irregularities such as unusually aggregated passenger counts, formatting issues, or inconsistent airline reporting. Autoencoders treat such deviations the same way they treat real-world anomalies, offering a useful mechanism to detect data-quality issues.



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8. Conclusion and Future Work

The project demonstrated the successful application of machine learning, clustering, and deep learning techniques to understand and forecast passenger traffic at SFO Airport. Regression models helped identify key operational drivers, while clustering revealed hidden structures in airline and terminal behaviour. Prophet effectively captured seasonal patterns and produced reliable forecasts. Autoencoders helped detect anomalies that align with major historical disruptions. Overall, the system built here provides actionable insights for airport operations, resource allocation, and long-term planning.

Future Work :

- Incorporate external variables such as weather, fuel prices, flight cancellations.
- Add holiday/event regressors to Prophet for more accurate forecasts.
- Build LSTM-based multivariate forecasting models.
- Use SHAP values for model explainability.
- Deploy as a web dashboard or API for real-time airport analytics.

9. References

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